Introduction to Artificial Intelligence (AI)

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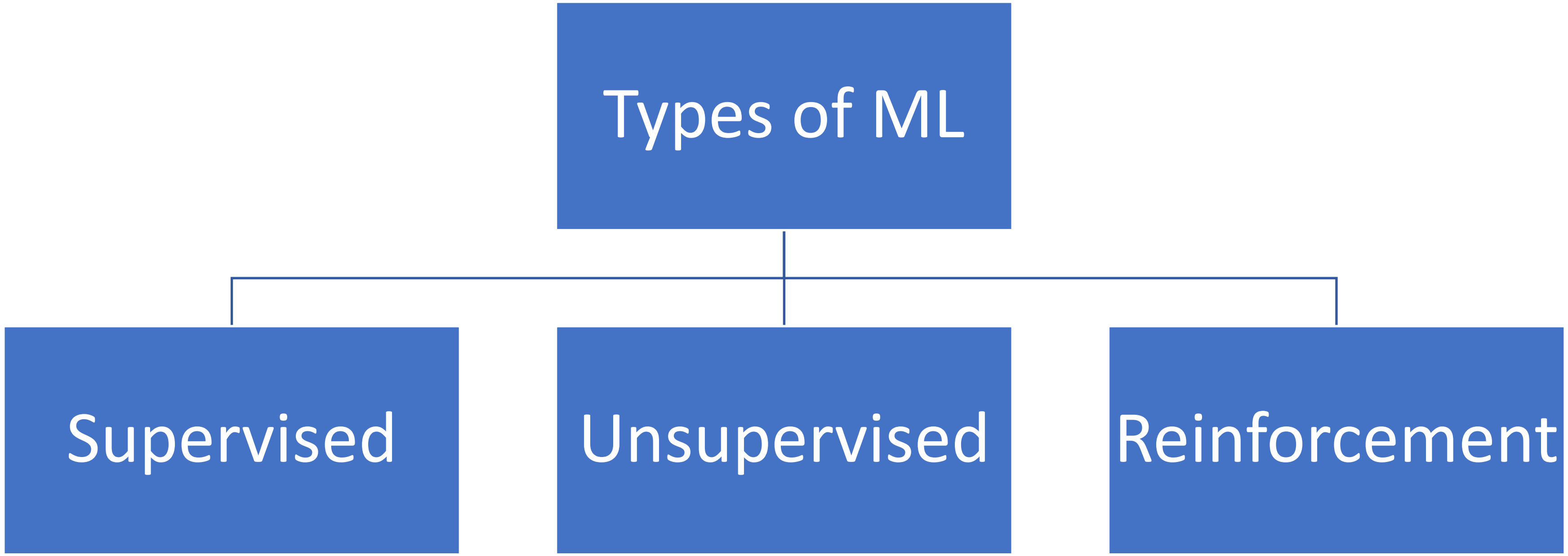
Contents

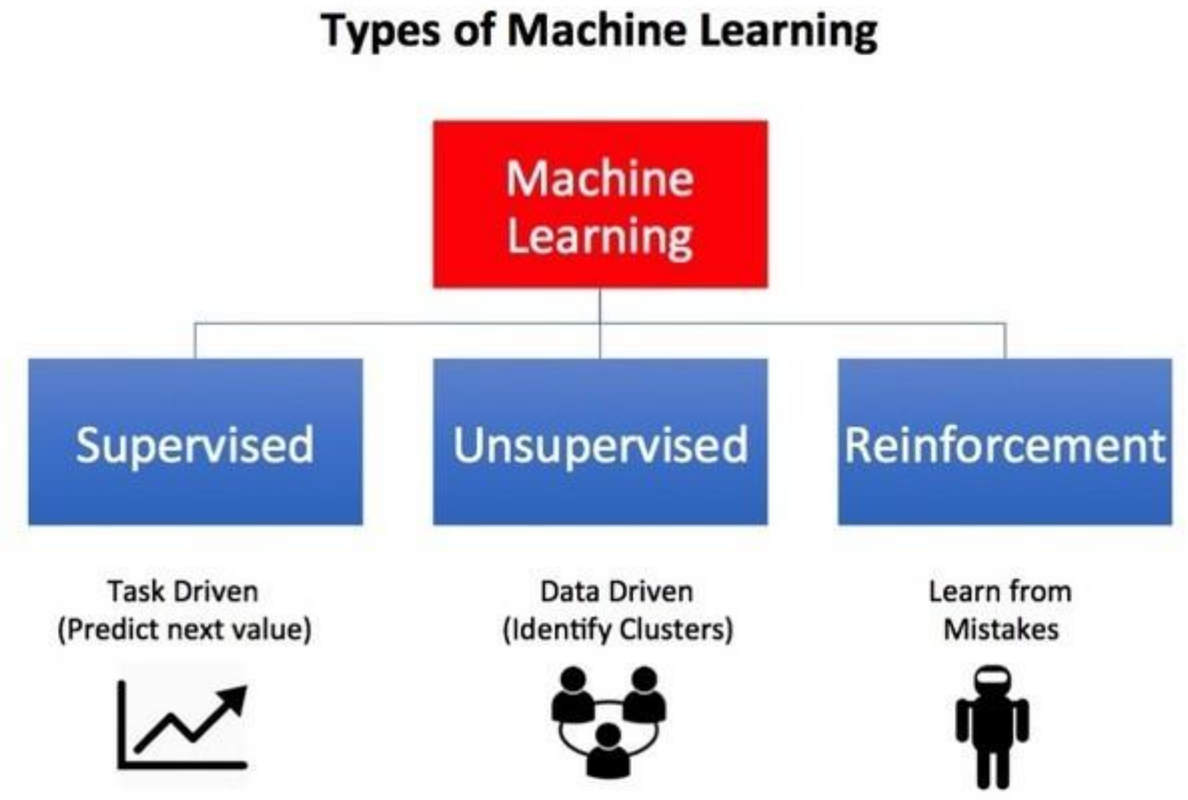
- Definition of AI : Definition of AI , What is agent, What is environment and Need of AI
- AI Understanding : What are AI Elements?
- Types of AI : Purely Reactive ,Limited Memory, Theory of Mind, Self aware
- Main Domains of AI technology : Data Science, Computer Vision and Natural Language Processing(NLP)
- History of AI
- Ways to implement AI : Introduction to Machine Learning and its categories of algorithm (Supervised and Unsupervised), Introduction to Deep Learning (input layer, hidden layer and output layer)
- AI Uses and its various Applications
- Advantages and Disadvantages of AI
- Introduction to Neural Networks : What is neural network? What is Fuzzy Logic? and what is the meaning of Genetic Algorithms?
- Current Trends and Future Directions in AI

Types of Machine Learning

- There are so many different types of Machine Learning systems that it is useful to classify them in broad categories, based on the following criteria
  - Whether or not they are trained with human supervision
    - supervised, unsupervised, and Reinforcement Learning



Supervised
Unsupervised
Reinforcement Learning



Supervised Learning

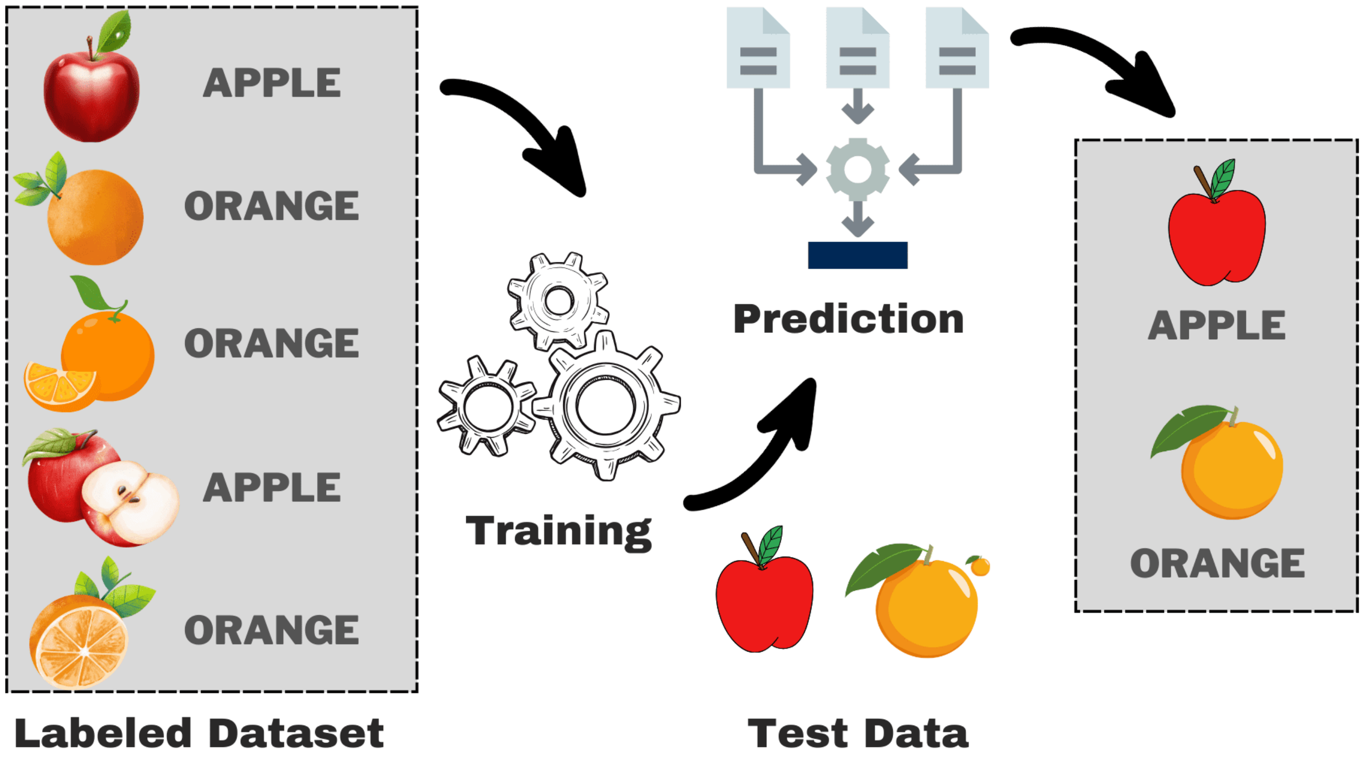
- The majority of practical machine learning uses supervised learning
- Supervised learning is where you have **input variables (x)** and an **output variable (Y)** and you use an algorithm to learn the mapping function from the input to the output

Y = f(X) ⇒ model / formula ,

- The goal is to approximate the mapping function so well that when you have new input data (x) that you can predict the output variables (Y) for that data
- It is called supervised learning because the process of an algorithm learning from the training dataset can be thought of as a teacher supervising the learning process
- We know the correct answers, the algorithm iteratively makes predictions on the training data and is corrected by the teacher
- Learning stops when the algorithm achieves an acceptable level of performance (measured in terms of **accuracy**)

Supervised Learning

- Output variable is already known for each input variable
- Algorithm learns to map input and output
- Model learns to associate features in the images with these predefine categories.



Supervised Learning – Problems

- **Regression**
  - Related to predicting future values
  - E.g.
    - Population growth prediction
    - Expecting life expectancy
    - Market forecasting/prediction
    - Advertising Popularity prediction
    - Stock prediction
  - Algorithms
    - Linear and multi-linear regression
    - Logistic regression
    - Naïve Bayes
    - Support Vector Machine

Supervised Learning – Problems

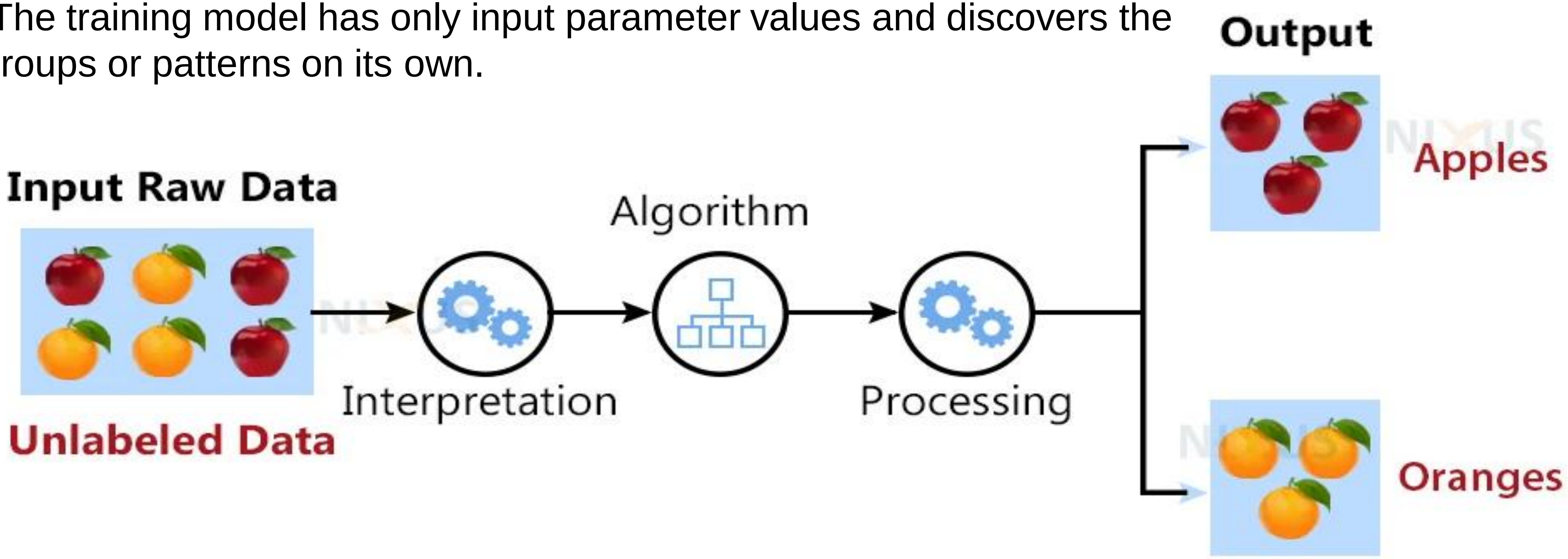
- **Classification**
  - Related to classify the records
  - Based on class / labels (eg. Email : Spam / Ham , Gender : Male / Female , Loan : Yes / No)
  - E.g.
    - Find whether an email received is a spam or ham
    - Identify customer segments
    - Find if a bank loan is granted
    - Identify if a kid will pass or fail in an examination
  - Algorithms
    - Logistic Regression
    - Decision Tree
    - Random Forest
    - Support Vector Machine
    - K-nearest neighbor

Unsupervised Learning

- Unsupervised learning is where you only have input data (X) and no corresponding output variables.
- The goal for unsupervised learning is to model the underlying structure or distribution in the data in order to learn more about the data.
- These are called unsupervised learning because unlike supervised learning above there is no correct answers and there is no teacher.
- Algorithms are left to their own devises to discover and present the interesting **structure** in the data.
- Structure in the form of GROUPS / CLUSTERS / ASSOCIATION
- Mostly used for EDA (Exploratory Data Analysis)

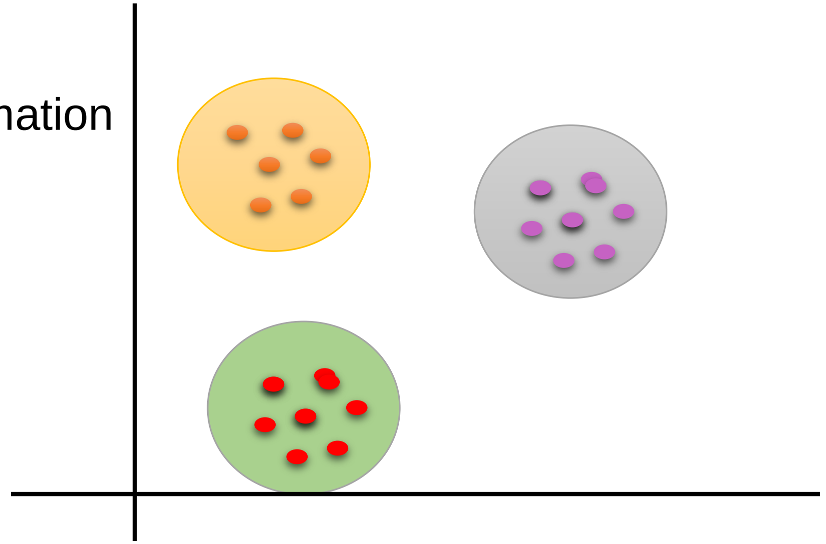
Unsupervised Learning

- Analyzes and clusters unlabeled datasets.
- These algorithms find hidden patterns and data without any human intervention.
- The training model has only input parameter values and discovers the groups or patterns on its own.



Unsupervised Learning - Problems

- **Clustering**
  - discover the inherent groupings in the data, such as grouping customers by purchasing behaviour
  - E.g.
    - Batsman vs bowler
    - Customer spending more money vs less money
    - Library
      - To cluster different books based on topics and information
  - Algorithms
    - K-means clustering
    - Hierarchical clustering



Unsupervised Learning - Problems

- **Association**
  - An association rule learning problem is where you want to discover rules that describe large portions of your data, such as people that buy X also tend to buy Y
  - prior knowledge of items/transactions
  - Items seen frequently in multiple transactions
  - E.g.
    - Market basket analysis
  - Algorithms
    - Apriori
    - Eclat

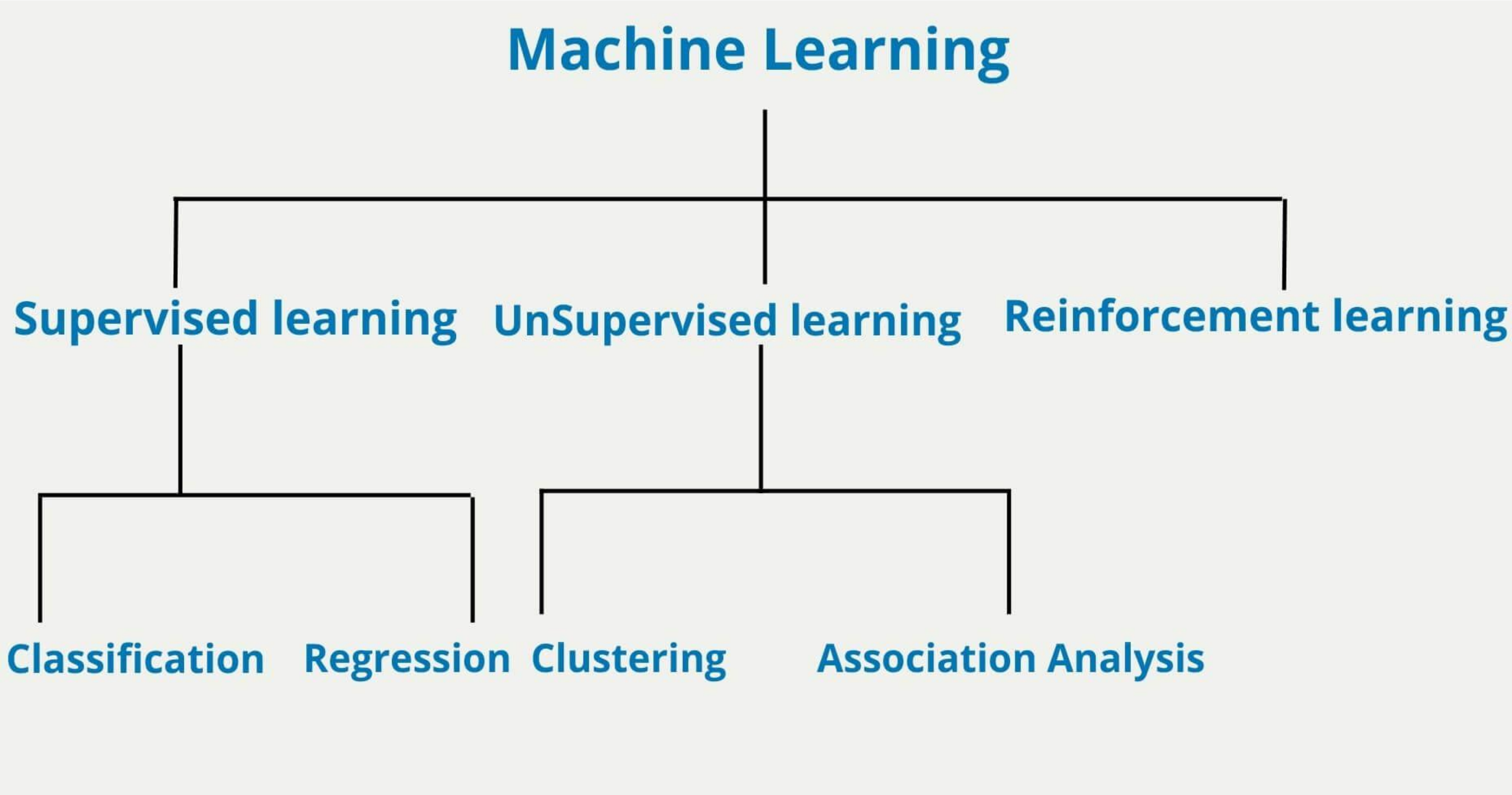
Market basket analysis

- Stationary items
- Cloth store
- Electronic gadgets

Reinforcement Learning

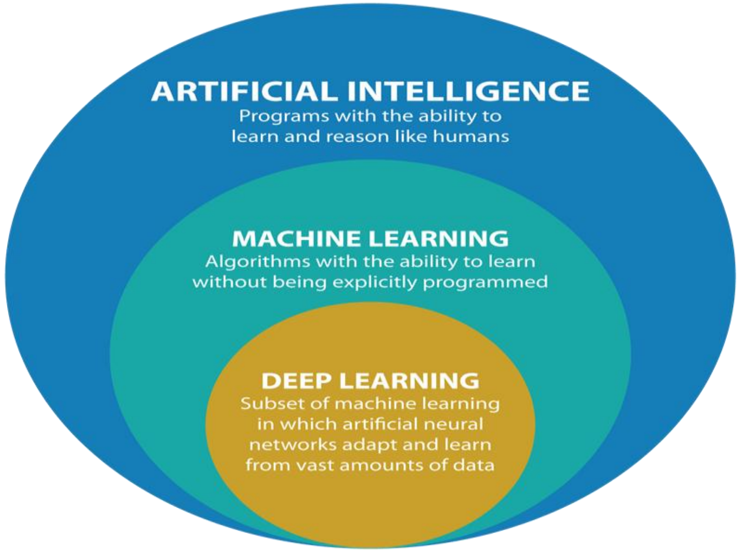
- It is about taking suitable action to maximize reward in a particular situation
- It is employed by various software and machines to find the best possible behaviour or path it should take in a specific situation
- Reinforcement learning differs from the supervised learning in a way that in supervised learning the training data has the answer key with it so the model is trained with the correct answer itself whereas in reinforcement learning, there is no answer but the reinforcement agent decides what to do to perform the given task
- In the absence of training dataset, it is bound to learn from its experience.
- Examples
  - Resources management in computer clusters
  - Traffic Light Control
  - Robotics
  - Web system configuration
  - Chemistry
- Algorithms
  - Q-Learning
  - Deep Q-Learning

Machine Learning



Deep Learning

- Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is inspired by the human brain.
- Deep learning models can recognize complex patterns in pictures, text, sounds, and other data to produce accurate insights and predictions.
- Artificial intelligence (AI) attempts to train computers to think and learn as humans do.
- Deep learning technology drives many AI applications used in everyday products, such as the following:
  - Digital assistants
  - Voice-activated television remotes
  - Fraud detection
  - Automatic facial recognition



Deep Learning

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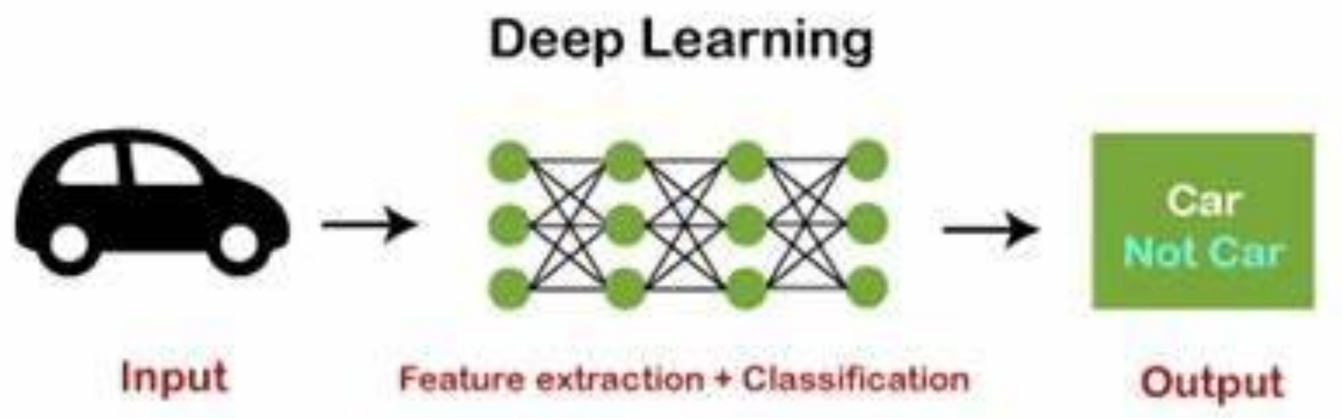
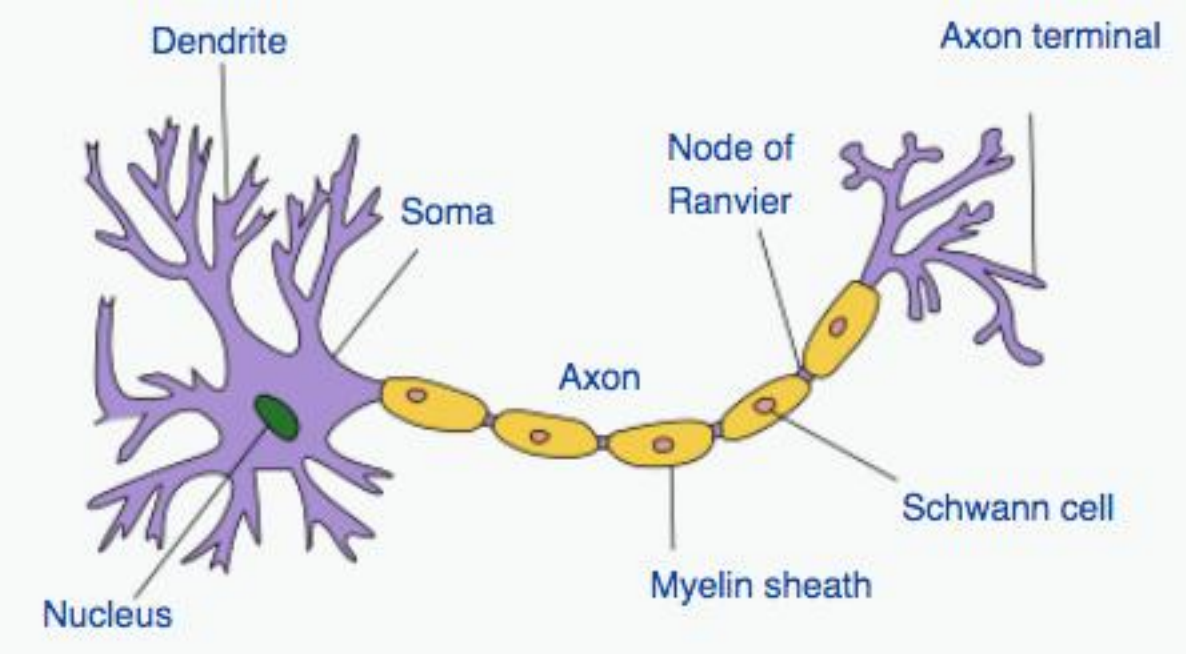
How does Deep Learning Works ?

- Deep learning algorithms are neural networks that are modeled after the human brain.
- For example, a human brain contains millions of interconnected neurons that work together to learn and process information.
- Similarly, deep learning neural networks, or artificial neural networks, are made of many layers of artificial neurons that work together inside the computer.



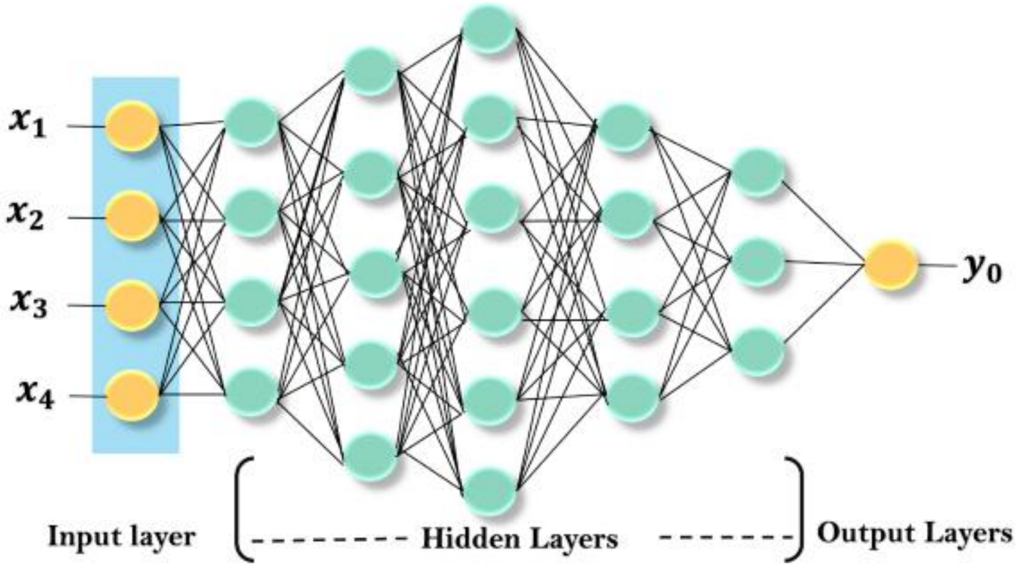
Neuron

- A neuron or nerve cell, is an electrically excitable cell that communicates with other cells via specialized connections called synapses
- A typical neuron consists of a cell body (soma), dendrites, and a single axon
- The soma is the body of the neuron
- The dendrites of a neuron are cellular extensions with many branches
- The axon primarily carries nerve signals away from the soma, and carries some types of information back to it



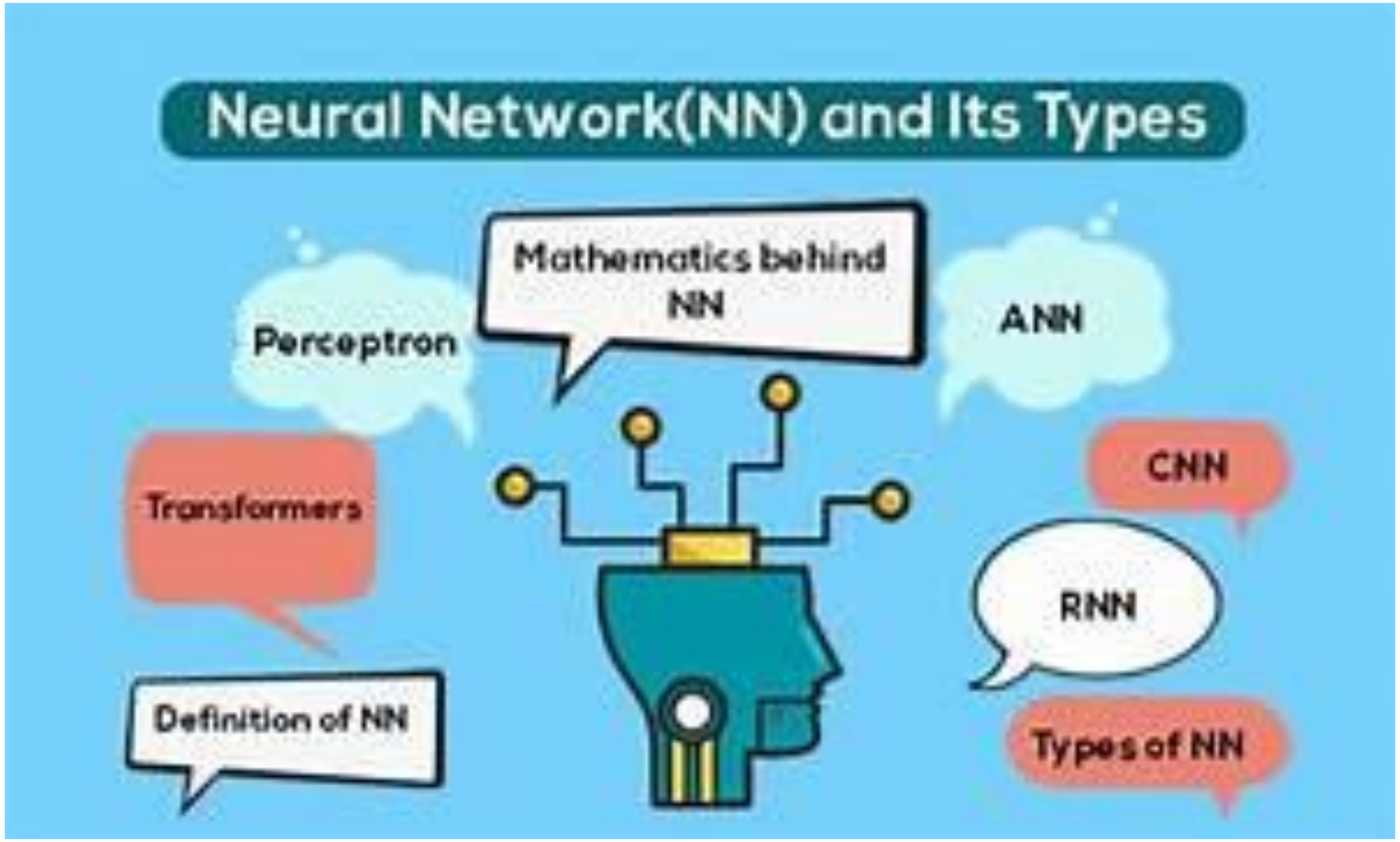
Neuron

- The first layer is called an Input layer, the last layer is called an output layer, and all layers between these two layers are called hidden layers.
- In the deep neural network, there are multiple hidden layers, and each layer is composed of neurons. These neurons are connected in each layer.
- The input layer receives input data, and the neurons propagate the input signal to its above layers.
- The hidden layers perform mathematical operations on inputs, and the performed data forwarded to the output layer.
- The output layer returns the output to the user.



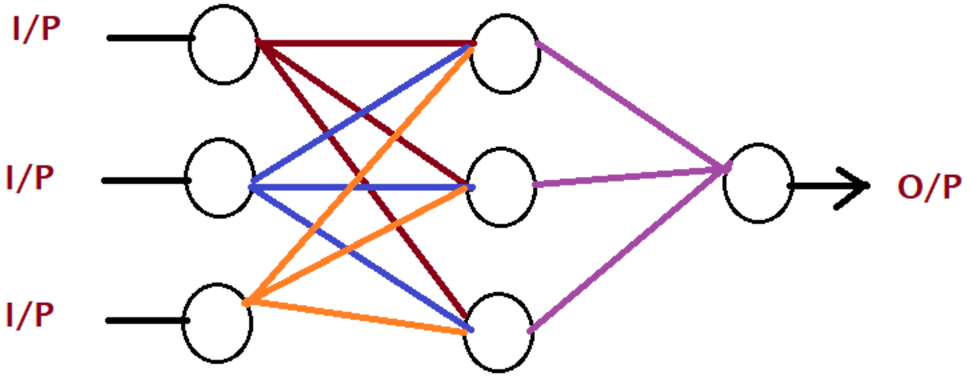
Types of Neural Network

- Artificial Neural Networks (ANN)
- Convolution Neural Networks (CNN)
- Recurrent Neural Networks (RNN)

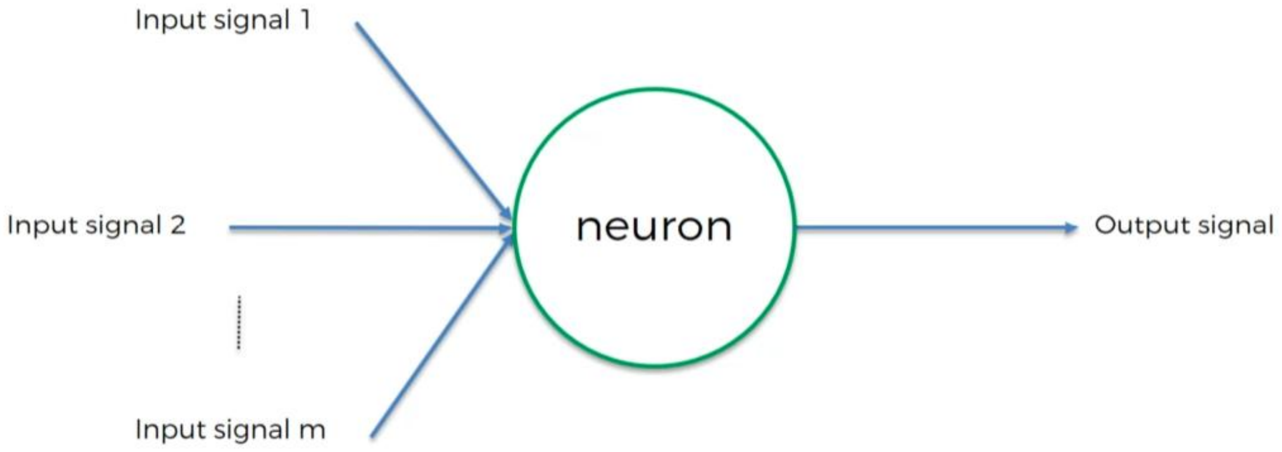


Artificial Neural Network

- Artificial neural networks (ANN) or connectionist systems are computing systems vaguely inspired by the biological neural networks that constitute animal brains
- The basic building block is a neuron



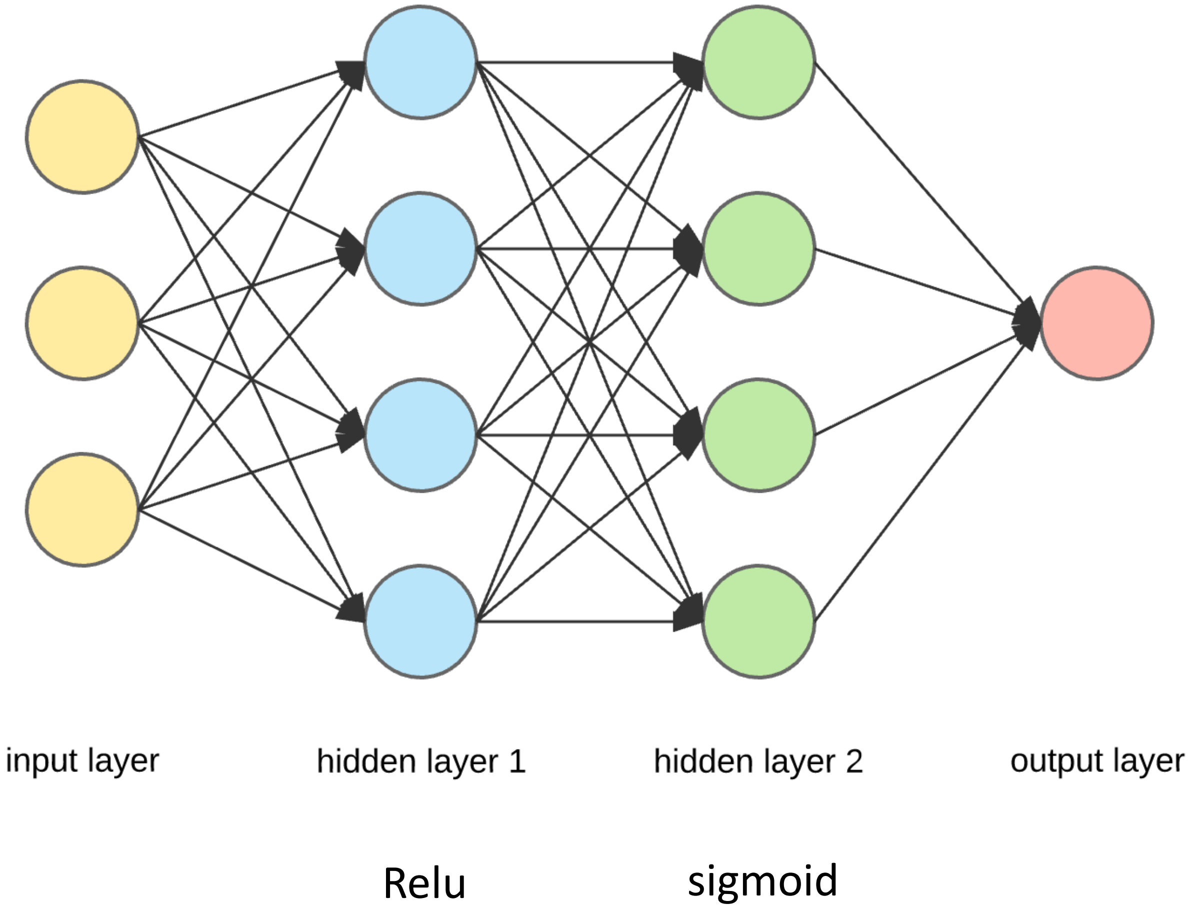
ANN



Artificial Neural Network

Artificial Neural Network

- Artificial Neural Networks (ANN) are **multi-layer** fully-connected neural nets
- They consist of an **input layer**, multiple hidden layers, and an **output layer**
- Every node in one layer is connected to every other node in the next layer
- We can make the network deeper by increasing the number of hidden layers.

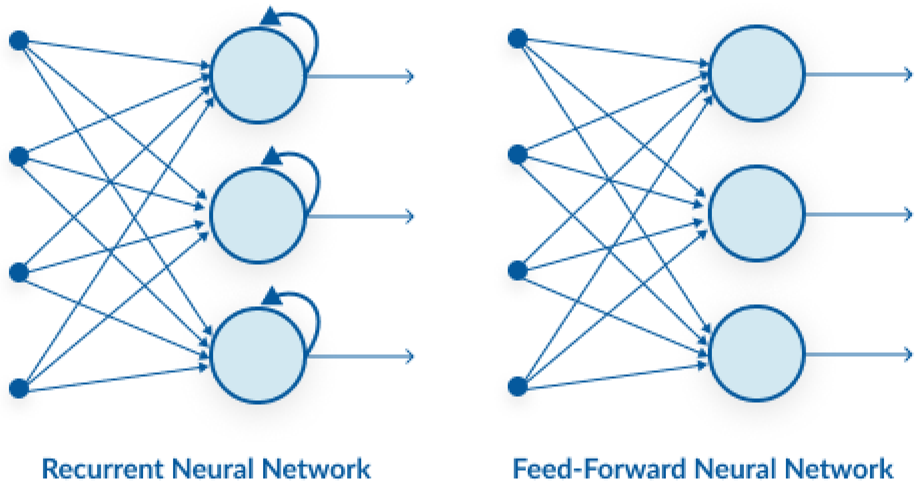


Characteristics of Artificial Neural Network

- It is neurally implemented mathematical model
- It contains huge number of interconnected processing elements called **neurons** to do all operations
- Information stored in the neurons are basically the weighted linkage of neurons
- The input signals arrive at the processing elements through connections and connecting weights.
- It has the ability to learn , recall and generalize from the given data by suitable assignment and adjustment of weights.
- The collective behaviour of the neurons describes its computational power, and no single neuron carries specific information

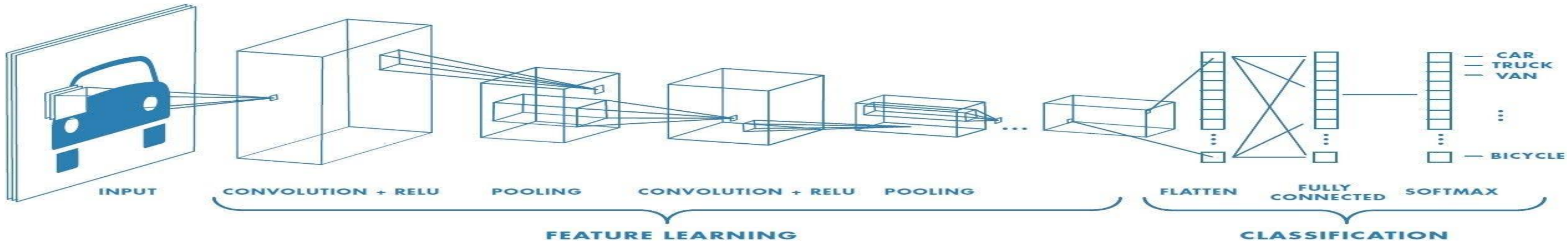
Recurrent Neural Networks(RNN)

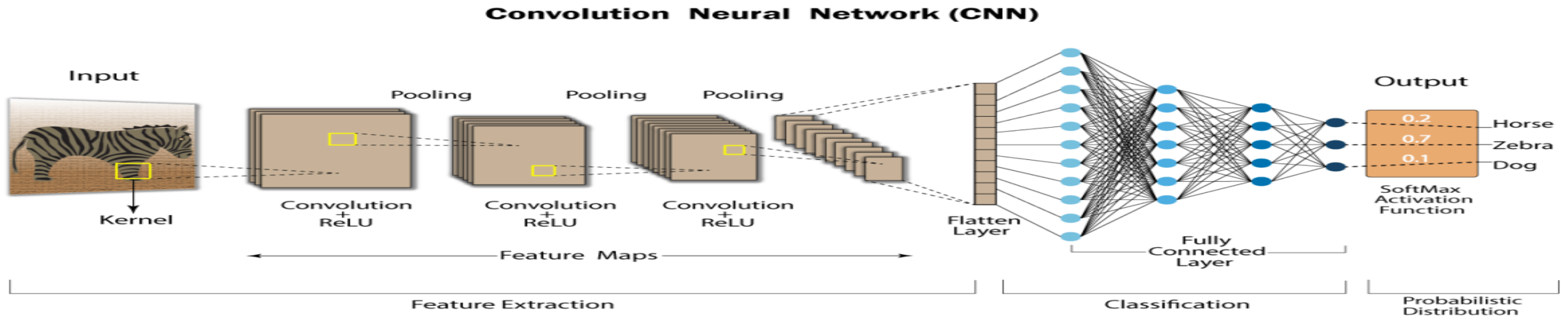
- A looping constraint on the hidden layer of ANN turns to RNN.
- We can use **Recurrent Neural Networks** to solve the problems related to:
  - Time Series data
  - Text data
  - Audio data



Convolution Neural Network (CNN)

- These CNN models are being used across different applications and domains, and they're especially used in image and video processing projects.





Application of Neural Network

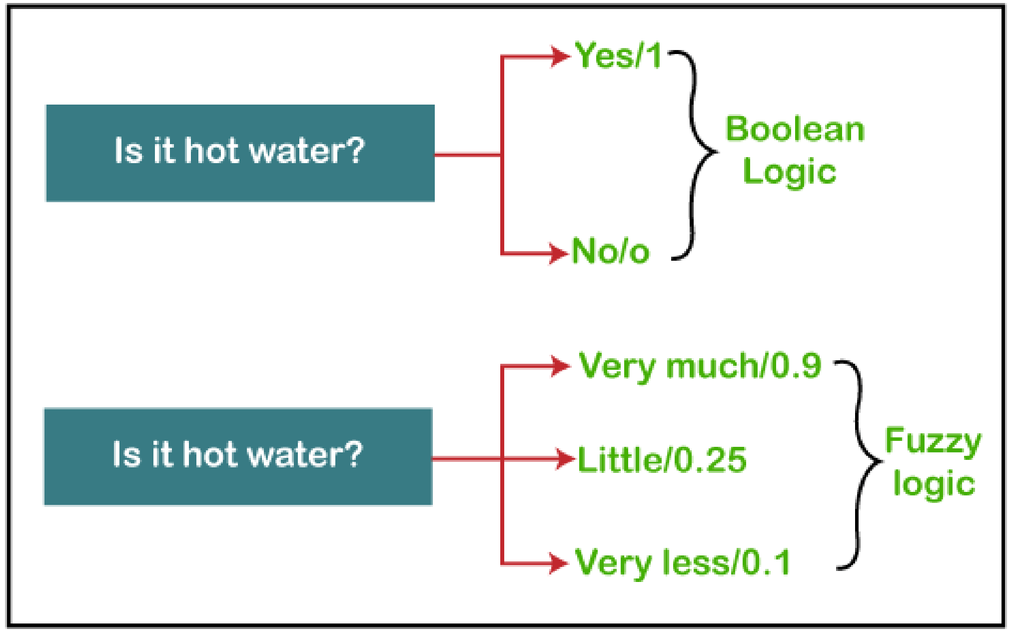
- Neural network is suitable for the research on *Animal behavior, predator/prey relationships etc.*
- It would be easier to do *proper valuation* of property, buildings, automobiles, machinery etc. with the help of neural network.
- Neural Network can be used in betting on horse races, sporting events and most importantly in stock market .
- It can be used to predict the correct judgement for any crime by using a large data of crime details as input and the resulting sentences as output.
- By analyzing data and determining which of the data has any fault (files diverging from peers) called as *Data mining, cleaning and validation* can be achieved through neural network.
- Neural Network can be used to predict targets with the help of echo patterns we get from sonar, radar, seismic and magnetic instruments .
- It can be used efficiently in *Employee hiring* so that any company can hire right employee depending upon the skills the employee has and what should be it's productivity in future .
- It has a large application in *Medical Research* .
- It can be used to for *Fraud Detection* regarding credit cards , insurance or taxes by analyzing the past records

Fuzzy Logic

- The **'Fuzzy'** word means the things that are not clear or are vague.
- Sometimes, we cannot decide in real life that the given problem or statement is either true or false.
- At that time, this concept provides many values between the true and false and gives the flexibility to find the best solution to that problem.
- In artificial intelligence systems, fuzzy logic is used to imitate human reasoning and cognition.
- fuzzy logic is well-suited for the following:
  - Engineering for decisions without clear certainties and uncertainties, or with imprecise data -- such as with natural language processing technologies; and
  - Regulating and controlling machine outputs, according to multiple inputs/input variables -- such as with temperature control systems.

Example of Fuzzy Logic

- Fuzzy logic contains the multiple logical values and these values are the truth values of a variable or problem between 0 and 1.
- This concept provides the possibilities which are not given by computers, but similar to the range of possibilities generated by humans.
- the fuzzy system, there are multiple possibilities present between the 0 and 1, which are partially false and partially true



AI Uses and Applications

Natural Language Processing (NLP)

- AI is used in NLP to analyze and understand human language.
- speech recognition, machine translation, sentiment analysis
- Eg. Siri , alexa

Image and Video Analysis

- AI techniques, including computer vision, enable the analysis and interpretation of images and videos.
- Eg. facial recognition, object detection, medical imaging

Robotics and Automation

- Robots equipped with AI algorithms can perform complex tasks in manufacturing, healthcare, logistics, and exploration.
- They can adapt to changing environments, learn from experience, and collaborate with humans.

AI Uses and Applications Cont…

Recommendation Systems

- AI-powered recommendation systems are used in e-commerce, streaming platforms, and social media to personalize user experiences.

Financial Services

- AI is extensively used in the finance industry for fraud detection, algorithmic trading, credit scoring, and risk assessment. Machine learning models can analyze vast amounts of financial data to identify patterns and make predictions.

Healthcare

- AI applications in healthcare include disease diagnosis, medical imaging analysis, drug discovery, personalized medicine, and patient monitoring. AI can assist in identifying patterns in medical data and provide insights for better diagnosis and treatment.

AI Uses and Applications Cont..

Virtual Assistants and Chatbots

- AI-powered virtual assistants and Chatbots interact with users, understand their queries, and provide relevant information or perform tasks. They are used in customer support, information retrieval, and personalized assistance.

Gaming

- AI algorithms are employed in gaming for creating realistic virtual characters, opponent behavior, and intelligent decision-making. AI is also used to optimize game graphics, physics simulations, and game testing.

Smart Homes and IoT

- AI enables the development of smart home systems that can automate tasks, control devices, and learn from user preferences. AI can enhance the functionality and efficiency of Internet of Things (IoT) devices and networks

Cybersecurity

- AI helps in detecting and preventing cyber threats by analyzing network traffic, identifying anomalies, and predicting potential attacks. It can enhance the security of systems and data through advanced threat detection and response mechanisms.

Advantages and Disadvantages of AI

- Pros
  - It reduces human error
  - It never sleeps, so it’s available 24x7
  - It never gets bored, so it easily handles repetitive tasks
  - It’s fast
- Cons
  - It’s costly to implement
  - It can’t duplicate human creativity
  - It will definitely replace some jobs, leading to unemployment
  - People can become overly reliant on it

Current Trends and Future Directions in AI

- When one considers the computational costs and the technical data infrastructure running behind artificial intelligence, actually executing on AI is a complex and costly
- AI in transportation
- AI in manufacturing
- AI in healthcare
- AI In education
- AI in media
- AI in customer service

"India's first AI mom, powered by real moms."

- **Meet Kavya Mehra — India's first AI-driven mom influencer**
- **Kavya is not just a digital persona but a unique blend of cutting-edge artificial intelligence and heartfelt storytelling, designed to connect with modern mothers in relatable and emotionally resonant ways.**

- Collective Artists Network's vision to merge technology with human creativity. "We wanted to explore the possibilities of combining AI with the emotional depth of real-life experiences. Kavya is designed to be a relatable voice for modern mothers while offering brands a way to create impactful, authentic content, she is AI-driven, it is easier for her to adapt real-time data, which makes her content continuously responsive to audience needs.
  Kavya's content based on trending conversations among real mothers, ensuring that she stays relevant to what's happening in their daily lives are regularly update.
- Designed to capture the essence of motherhood and engage with audiences in pretty much all aspects of motherhood.

Neural Networks

- A **Perceptron** is an **Artificial Neuron**.
- Perceptron is one of the simplest Artificial neural network architectures.
- $\sum wixi + bias = w1x1 + w2x2 + w3x3 + bias$
- output = $f(x) = 1$ if $\sum w1x1 + b >= 0$; 0 if $\sum w1x1 + b < 0$
- **b = Bias**: The bias term helps the perceptron make adjustments independent of the input, improving its flexibility in learning.
- **w = Weights**: Each input feature is assigned a weight that determines its influence on the output. These weights are adjusted during training to find the optimal values.

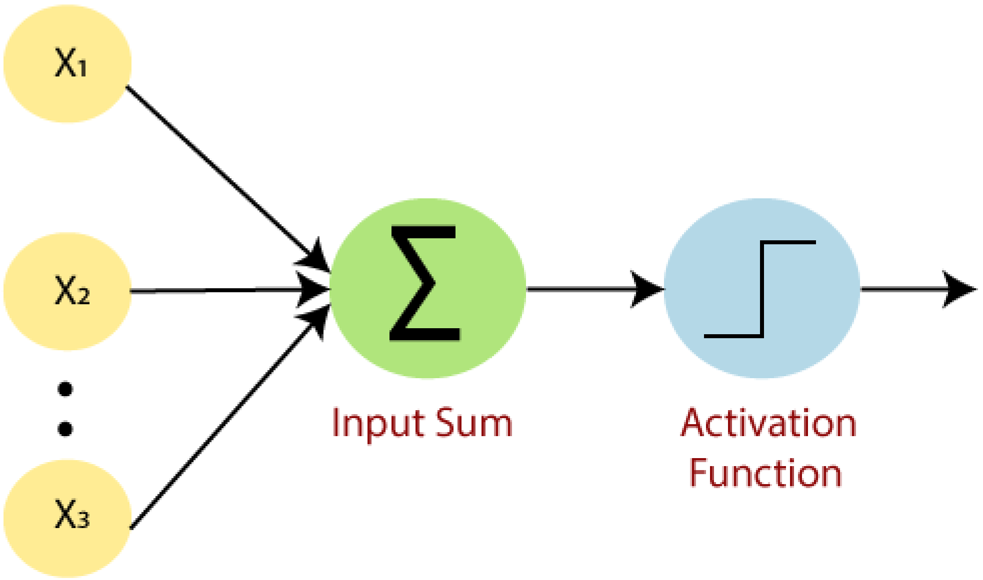
$$y = \begin{cases} 0, & \text{if } w \cdot x + b \leq 0 \\ 1, & \text{if } w \cdot x + b > 0 \end{cases}$$

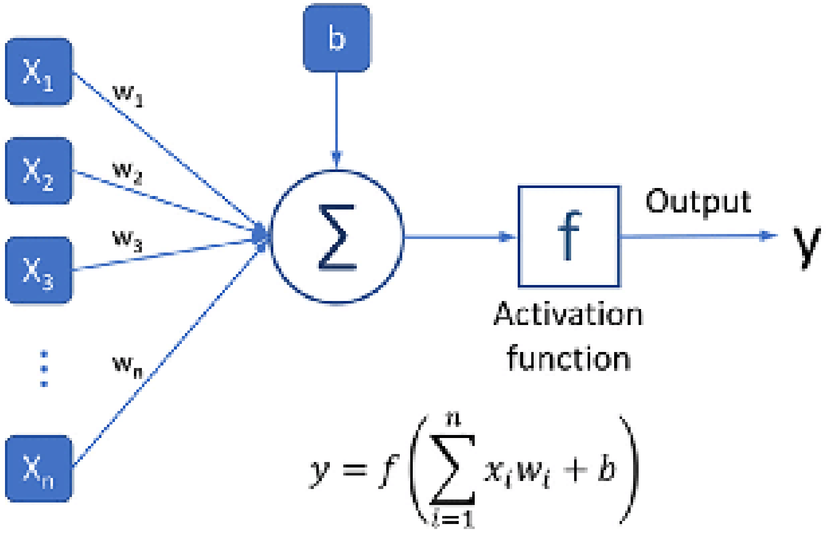
Components of Perceptrons

- **b = Bias:** The bias term helps the perceptron make adjustments independent of the input, improving its flexibility in learning.
- **w = Weights:** Each input feature is assigned a weight that determines its influence on the output. These weights are adjusted during training to find the optimal values.
- **Summation Function**: The perceptron calculates the weighted sum of its inputs using the summation function. The summation function combines the inputs with their respective weights to produce a weighted sum.
- **Activation Function**: The weighted sum is then passed through an activation function, which take the summed values as input and compare with the threshold and provide the output as 0 or 1.
- **Output:** The final output of the perceptron, is determined by the activation function’s result.
- For example, in binary classification problems, the output might represent a predicted class (0 or 1).

Neural Networks

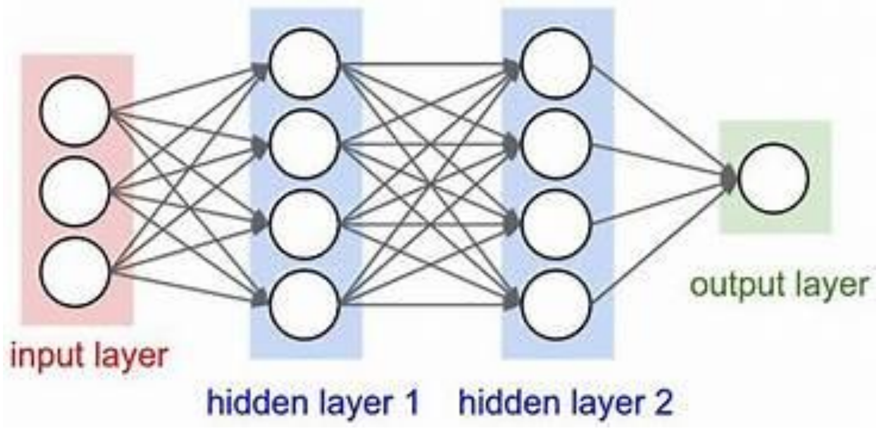
- **Single-Layer Perceptron**
- A single-layer perceptron can solve problems that are linearly separable by learning a linear decision boundary.
- Single-layer perceptron adequate for solving the OR AND logical gate problem.
- They are effective for tasks where the data can be divided into distinct categories through a straight line





Neural Networks

- **Multi-layer Perceptron**
- A multi-layer neural network which has multiple layer of neurons such as input layer, hidden layer, and output layer.
- The XOR (exclusive OR) is a simple logic gate problem that cannot be solved using a single-layer perceptron (a basic neural network model).
- Because XOR is not linearly separable, no single line (or hyperplane) can separate the outputs 0 and 1.
- Thus making a single-layer perceptron inadequate for solving the XOR problem.
- It can handle more complex patterns and relationships within the data



Activation Function

- **Sigmoid Function** :-
  - The sigmoid activation function is a mathematical function that maps any real-valued number into a value between 0 and 1. The **output is always between 0 and 1,**
  - It's commonly used in binary classification tasks.

- **Relu Function (Rectified Linear Unit)** :-
  - The ReLU activation function is one of the most widely used activation function.
  - The **output ranges from 0 to infinity**.
  - ReLU is commonly used in hidden layers of deep neural networks

Machine Learning Vs Deep Learning

Machine Learning	Deep Learning
Used when data is less and structured	Used when data is huge and unstructured
Traditional machine learning methods require human input for the machine learning software to work sufficiently well	in deep learning, the data scientist gives only raw data to the software
A data scientist manually determines the set of relevant features that the software must analyze	The deep learning network derives the features by itself and learns more independently
▪ This limits the software's ability, which makes it tedious to create and manage	It can analyze unstructured datasets like text documents, identify which data attributes to prioritize, and solve more complex problems

Thank You!!